Machine Learning-Based ICU Mortality Prediction Under Class Imbalance and Label Noise

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Abstract

Predicting mortality risk in intensive care units (ICUs) is an important clinical task because critically ill patients may deteriorate rapidly and require timely escalation of care. Accurate early prediction can support closer monitoring, earlier intervention, and more effective allocation of limited hospital resources. Machine learning offers a data-driven alternative to traditional severity scores, but healthcare datasets may contain class imbalance, imperfect labels, and documentation errors that reduce model reliability. This study developed a machine learning framework for ICU mortality prediction using a public structured dataset containing 15,000 patient records and 24 variables. The workflow included preprocessing, feature refinement, class balancing, comparative evaluation of eight classifiers, threshold optimization, and robustness testing under simulated label noise. A clinically interpretable seven-feature subset was retained for final modeling. Among the evaluated approaches, a tuned XGBoost model with a probability threshold of 0.35 provided the most practical balance for the study objective, achieving a recall of 0.885, F1-score of 0.678, and ROC-AUC of 0.641. Additional testing under 15% label noise highlighted the importance of robustness assessment in clinical prediction tasks. Feature importance analysis identified vasopressor use, ventilation requirement, and sepsis status as the strongest mortality indicators. The findings suggest that recall-focused and interpretable machine learning models can support ICU risk stratification under imperfect data conditions.

1 Introduction and Motivation

Timely identification of patients at high risk of death remains a major challenge in intensive care units (ICUs). Critically ill patients may deteriorate rapidly, and delays in recognizing worsening conditions can reduce opportunities for early intervention, closer monitoring, or escalation of treatment. Because ICU resources such as beds, ventilators, specialist staff, and advanced therapies are limited, hospitals increasingly rely on risk assessment methods to support both clinical prioritization and operational planning.

Traditional scoring systems such as the Acute Physiology and Chronic Health Evaluation (APACHE) and Sequential Organ Failure Assessment (SOFA) are widely used to estimate illness severity and mortality risk. These tools remain clinically valuable because they provide standardized measures based on patient physiology and organ dysfunction. However, conventional scores are generally derived from predefined statistical relationships and may not fully capture complex nonlinear interactions among demographic factors, comorbidities, treatment requirements, and rapidly changing patient conditions (Zhang et al., 2025; Kim et al., 2022).

Machine learning (ML) has emerged as a promising complementary approach for clinical outcome prediction because it can learn predictive patterns directly from data and capture relationships that may be difficult to represent using traditional methods (Esteva et al., 2020). Recent studies have reported encouraging results for ICU mortality prediction, sepsis outcome estimation, deterioration

alerts, and decision-support systems using algorithms such as logistic regression, random forests, support vector machines, boosting methods, and ensemble learning (Machado et al., 2020; Fadlallah et al., 2021; Shashikumar et al., 2023). Despite this progress, practical deployment remains challenging. Healthcare datasets may contain class imbalance, noisy labels, incomplete documentation, and population differences that reduce model reliability when applied in real settings (Li et al., 2023; Cabitza et al., 2022).

Motivated by these challenges, this study develops an interpretable machine learning framework for ICU mortality prediction using a public structured dataset containing 15,000 patient records. Rather than focusing only on overall accuracy, the framework emphasizes clinically relevant model selection through recall, F1-score, ROC-AUC, threshold optimization, and robustness testing under simulated label noise. The objective is to determine whether a practical prediction pipeline can identify high-risk ICU patients while maintaining transparency, balanced performance, and awareness of imperfect medical data conditions.

2 Related Work

Applications of machine learning in critical care have expanded rapidly in recent years, particularly for mortality estimation, deterioration detection, and treatment-support systems. Prior literature indicates that no single algorithm consistently dominates across all ICU settings, since performance often depends on dataset quality, cohort definition, variable selection, outcome prevalence, and evaluation priorities.

2.1 ICU Mortality Prediction Models

Several papers have examined ICU mortality estimation using structured clinical data, with ensemble and tree-based approaches frequently reporting strong results. Kim et al. (2022) developed models for both mortality and length-of-stay forecasting, showing that richer variable sets can improve performance beyond conventional severity scoring alone. Using COVID-19 ICU data, Fadlallah et al. (2021) applied a SuperLearner stacking framework, illustrating the value of combining multiple learners in high-acuity settings. Li et al. (2023) focused on cardiac-arrest ICU patients using MIMIC-IV, while Khan et al. (2021) compared several classifiers on broader ICU cohorts. Collectively, these papers suggest that model success is highly dependent on patient population and dataset composition, although boosting and ensemble methods often remain competitive.

2.2 Sepsis and High-Risk Clinical Complications

Sepsis remains one of the most studied ICU complications because delayed recognition can significantly worsen outcomes. In a systematic review, Shashikumar et al. (2023) found that machine learning methods often improved early sepsis detection compared with rule-based systems, especially when electronic health record variables were incorporated. Liu et al. (2024) reported useful prognostic accuracy for sepsis-related mortality, while Wang et al. (2024) showed predictive value in patients with sepsis-associated acute kidney injury. These results suggest that targeted subgroup models may capture condition-specific risk patterns more effectively than one general model for all ICU patients.

2.3 Early Warning and Clinical Decision Support

Beyond mortality estimation, researchers have emphasized proactive warning systems and operational decision support. Machado et al. (2020) validated a multicenter early warning model across Brazilian hospitals, indicating that predictive systems can generalize when externally tested. Reviewing broader ICU applications, Sendak et al. (2021) concluded that machine learning may assist with triage support, deterioration alerts, and workflow prioritization. However, both papers emphasized that real clinical value depends on usability, trust, and workflow integration rather than accuracy alone.

2.4 Reliability, Deployment, and Research Gap

Although reported metrics are often promising, implementation barriers remain substantial. Komorowski et al. (2024) discussed obstacles such as data heterogeneity, workflow disruption, and limited external validation. Cabitza et al. (2022) similarly identified weak transparency, limited clinician trust, and adoption challenges in ICU settings. In addition, Chen et al. (2023) examined risk modelling under changing demographics, highlighting the importance of maintaining performance when data distributions shift over time.

Table 1: Representative Prior Studies in ICU Mortality Prediction and Clinical Decision Support

Study	Dataset / Scope	Methods	Main Findings	Limitation
Fadlallah et al. (2021)	COVID ICU patients	SuperLearner	Strong mortality estimation	Disease-specific population
Kim et al. (2022)	ICU length of stay and mortality	ML ensemble methods	Good predictive utility	Generalizability concerns
Li et al. (2023)	Cardiac arrest ICU patients from MIMIC-IV	Multiple ML models	Strong in-hospital mortality prediction	Narrow subgroup
Liu et al. (2024)	Sepsis ICU patients	ML classifiers	Useful mortality estimation	Disease-specific
Sendak et al. (2021)	ICU clinical decision support review	Literature review	ML is promising for ICU decision support	Implementation barriers

Overall, prior work demonstrates the promise of machine learning in ICU outcome tasks, but fewer studies jointly address class imbalance, threshold selection, interpretability, and robustness to noisy labels within one consistent framework. This gap motivates the present study, which compares multiple classifiers, applies feature refinement, prioritizes recall-sensitive model selection, and evaluates performance under simulated label noise conditions.

3 Data Description and Preprocessing

3.1 Dataset Description

This study used a publicly available ICU Patient Monitoring and Mortality Prediction dataset obtained from Kaggle. The dataset contains 15,000 patient records and 24 structured variables representing demographic information, admission context, physiological measurements, treatment interventions, severity indicators, and the final mortality outcome. The target variable was `mortality_label`, where 0 indicates survived and 1 indicates died.

The dataset showed moderate class imbalance, with survival cases representing the majority class and mortality cases forming a smaller proportion. Available predictors included age, gender, admission type, comorbidity burden, heart-rate statistics, blood-pressure measures, respiratory rate, oxygen saturation, temperature, glucose, lactate, urine output, ventilation requirement, vasopressor use, APACHE score, SOFA score, sepsis flag, and length of stay.

3.2 Data Cleaning and Preparation

Identifier fields such as `patient_id` were removed because they do not contain predictive clinical information and could introduce artificial patterns unrelated to patient risk. The dataset was reviewed for missing values, variable consistency, and data types. Categorical variables were converted into numerical format for model compatibility, while binary indicators were retained in interpretable 0/1 form. Continuous features with different ranges were standardized inside model pipelines when algorithms were sensitive to scale, particularly support vector machines and K-nearest neighbors.

To prevent data leakage, preprocessing transformations were embedded inside machine learning pipelines and applied only after train-test splitting. This ensured that information from the test set did not influence training-stage preprocessing. The training partition was then used for balancing and model development, while the test partition remained untouched for final evaluation.

3.3 Exploratory Data Analysis

Initial exploratory analysis was conducted to understand the behavior of the dataset and identify patterns relevant to mortality prediction. Class distribution analysis confirmed moderate imbalance between survivors and non-survivors, with survival cases representing the majority outcome class and mortality cases forming a substantially smaller proportion. This imbalance highlighted the importance of applying balancing strategies during later model development, as shown in Figure 1.

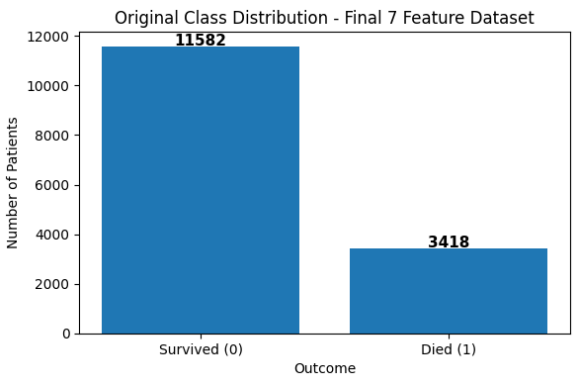

Figure 1: Class distribution of mortality outcomes in the refined seven-feature ICU dataset before balancing.

Correlation heatmaps and summary comparisons showed that severity-related variables such as APACHE score, SOFA score, ventilation requirement, vasopressor use, and sepsis status were more strongly associated with mortality outcomes than many routine physiological variables. Boxplots and grouped comparisons suggested that nonsurvivors generally presented higher illness severity and greater treatment intensity. However, some variables showed weaker separation than expected, indicating that ICU mortality is influenced by multiple interacting factors rather than any single measurement.

3.4 Feature Refinement and Class Balancing

Following exploratory analysis and practical clinical reasoning, a refined subset of seven predictors was selected for the strongest final experiments: `ventilation_required`, `vasopressor_used`, `sepsis_flag`, `apache_score`, `sofa_score`, `comorbidity_score`, and `age`. This reduced feature set captures treatment intensity, organ dysfunction, baseline vulnerability, and overall severity while improving interpretability compared with broader feature collections.

To address the original class imbalance, random undersampling was applied to the training data during the final modeling stage. This helped reduce majority-class dominance and improved the model's ability to detect mortality cases more effectively. Although undersampling may remove some majority-class information, it provided a practical and computationally efficient strategy for fairer classifier training in this project.

4 Methodology

4.1 Experimental Pipeline

The overall workflow followed a structured machine learning pipeline designed for reliable healthcare prediction. The main stages included dataset preparation, exploratory analysis, preprocessing, baseline model comparison, hyperparameter optimization, class imbalance handling, threshold optimization, robustness testing, and final interpretation. Figure 2 illustrates the complete methodology used in this study.

The pipeline was intentionally designed to move beyond simple model training by incorporating threshold optimization, class imbalance handling, and robustness assessment under imperfect labels. This reflects real healthcare settings, where data quality and evaluation priorities may differ from ideal benchmark conditions.

4.2 Train-Test Strategy and Validation

The dataset was divided into training and testing subsets to evaluate generalization performance on unseen data. Model development steps, including preprocessing and parameter tuning, were conducted using the training data only, while the held-out test set was reserved for final evaluation. Cross-validation procedures were used for baseline assessment and hyperparameter tuning to reduce the chance of selecting a model based on one favorable split.

4.3 Baseline Classification Models

Eight supervised learning algorithms were evaluated as baseline candidates: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, Naive Bayes, Gradient Boosting, and XGBoost. These models were selected because they represent linear, probabilistic, distance-based, tree-based, and ensemble learning approaches commonly used in healthcare prediction research.

4.4 Hyperparameter Tuning and Threshold Optimization

After baseline comparison, the strongest models underwent further optimization using GridSearchCV. Hyperparameters such as tree depth, number of estimators, learning rate, regularization strength, and kernel settings were tuned depending on the algorithm. Most binary classifiers use a default probability threshold of 0.50, but in ICU mortality prediction, failing to identify a truly high-risk patient may be more harmful than generating additional false alarms. For this reason, multiple thresholds were tested for the strongest candidate models. A threshold of 0.35 was selected for the final tuned XGBoost model because it achieved the most practical balance between recall, F1-score, and acceptable discrimination performance.

4.5 Label Noise Robustness Test

Healthcare datasets may contain coding mistakes, delayed documentation, or uncertain outcome labels. To simulate imperfect labels, an additional robustness experiment introduced 15% random label flipping within the training data. The purpose was not to create a better final model, but to observe

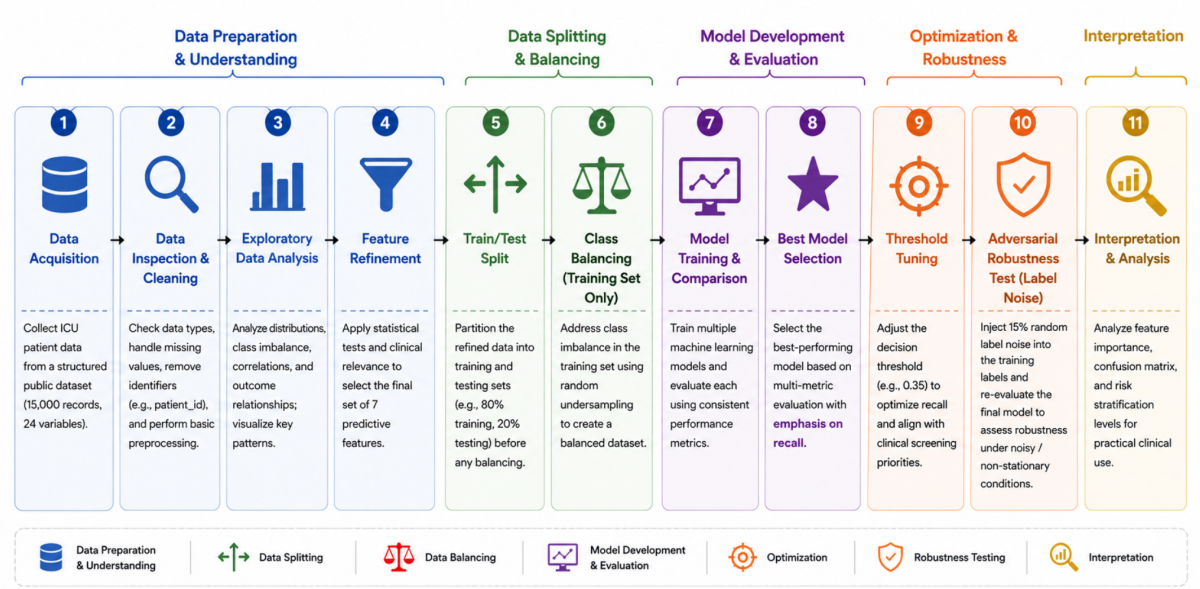

Figure 2: Proposed machine learning workflow for ICU mortality prediction under class imbalance and label noise conditions.

how prediction behavior changed when training labels became less reliable. This provides a practical stress test because real clinical datasets are rarely error-free.

4.6 Evaluation Metrics and Interpretability

Models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Recall was given particular attention because identifying high-risk mortality cases was a central project objective. To improve interpretability, the final model was also analyzed using feature importance rankings and probability-based risk grouping.

5 Experimental Results and Discussion

5.1 Baseline Model Comparison

Eight supervised learning algorithms were first evaluated using the refined, balanced dataset. Performance was assessed using accuracy, precision, recall, F1-score, and ROC-AUC to provide a broader view than relying on a single metric. The results showed that overall model performance was relatively close across several classifiers, indicating that the selected seven predictors carried useful mortality information regardless of model family. However, clearer differences appeared when clinically important measures such as recall and F1-score were considered.

Among the baseline models, support vector machines, random forest, gradient boosting, and XGBoost produced stronger overall balance than simpler decision tree or K-nearest neighbors approaches. Logistic regression remained competitive and provided a transparent linear benchmark, while Naive Bayes achieved reasonable discrimination despite its simplifying independence assumptions. These findings are consistent with prior healthcare studies reporting that ensemble and margin-based models often outperform simpler methods when predictor relationships are nonlinear or interactive (Kim et al., 2022; Esteva et al., 2020; Liu et al., 2024).

5.2 Effect of Threshold Optimization

Although several baseline models achieved similar ROC-AUC values, threshold adjustment substantially changed practical usefulness for the strongest candidate models. After baseline comparison, the tuned XGBoost classifier was further evaluated using three probability thresholds: 0.35, 0.40, and 0.45. Among these tested settings, the 0.35 threshold produced the strongest practical balance for the project objective, achieving the highest recall of 0.885 while maintaining a competitive F1-score and acceptable overall discrimination. Higher thresholds reduced false positive predictions, but they also missed more mortality cases.

5.3 ROC Analysis and Classification Behavior

ROC curve analysis showed moderate but stable discrimination across several leading models. Gradient Boosting achieved the highest ROC-AUC of 0.649, followed closely by Naive Bayes, XGBoost, Random Forest, and Logistic Regression. The rel-

Table 2: Comparative Performance of Evaluated Models on the Test Set

Model	Threshold	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Tuned XGBoost	0.35	0.5804	0.5500	0.8845	0.6783	0.6415
Tuned XGBoost	0.40	0.5885	0.5622	0.7997	0.6602	0.6415
Tuned XGBoost	0.45	0.6016	0.5836	0.7091	0.6403	0.6415
SVM	0.50	0.5965	0.5912	0.6257	0.6080	0.6335
Random Forest	0.50	0.6009	0.5977	0.6170	0.6072	0.6411
XGBoost	0.50	0.5987	0.5971	0.6067	0.6019	0.6415
Gradient Boosting	0.50	0.5980	0.5968	0.6038	0.6003	0.6494
Naive Bayes	0.50	0.6038	0.6102	0.5746	0.5919	0.6458
Logistic Regression	0.50	0.6075	0.6184	0.5614	0.5885	0.6407
KNN	0.50	0.5673	0.5665	0.5731	0.5698	0.5818
Decision Tree	0.50	0.5768	0.5827	0.5409	0.5610	0.6132

atively narrow AUC range suggests that predictor quality may have influenced performance more than model choice alone. Once clinically relevant predictors were selected, multiple model families were able to learn useful decision boundaries.

Figure 3 presents the ROC curves for evaluated classifiers together with the confusion matrix of the final tuned XGBoost model. The confusion matrix further demonstrated its recall-focused behavior, with substantially more correctly detected mortality cases than missed deaths.

5.4 Interpretability and Clinical Relevance

Feature importance analysis showed that vasopressor use, ventilation requirement, and sepsis status were the strongest contributors to mortality prediction, followed by APACHE score, comorbidity burden, SOFA score, and age. These findings are clinically plausible. Patients requiring vasopressors or mechanical ventilation are commonly among the most unstable ICU cases, while sepsis remains a major contributor to critical deterioration. Severity scores and age also remain established mortality indicators. This agreement between model behavior and clinical expectations improves confidence that the system learned relevant patterns rather than random statistical noise.

As shown in Figure 4, treatment intensity indicators and severity-related variables contributed most strongly to the final tuned XGBoost model.

5.5 Label Noise Robustness Test

To examine reliability under imperfect data conditions, an additional experiment introduced 15% random label noise during training. Under this stress condition, recall increased further, but precision and overall balance deteriorated. This behavior is informative rather than purely negative. It suggests that noisy labels can distort the decision boundary and encourage excessive positive predictions. In healthcare settings, such instability may create unnecessary alarms and reduce trust in automated systems.

Table 3: Impact of Label Noise on Final Tuned XGBoost Performance

Model Version	Acc.	Prec.	Rec.	F1	AUC
Clean final XGBoost (0.35)	0.5804	0.5500	0.8845	0.6783	0.6415
XGBoost + 15% label noise	0.5373	0.5197	0.9839	0.6801	0.6481

5.6 Overall Discussion

Across all experiments, the strongest final solution was not the model with the highest standalone accuracy, but the model offering the best balance between mortality detection, interpretability, and robustness. The tuned XGBoost model with threshold optimization addressed these priorities most effectively. The results also demonstrate that careful preprocessing, balanced training, clinically guided feature refinement, and objective-based threshold tuning can be as influential as algorithm choice itself. For practical ICU deployment, these findings support machine learning as a decision-support aid for identifying patients who may require earlier review or closer monitoring, rather than as a replacement for physician judgment.

6 Conclusion and Future Work

This study developed and evaluated a machine learning framework for ICU mortality prediction using a structured public dataset containing 15,000 patient records. The project examined multiple supervised learning algorithms under a consistent experimental pipeline that included preprocessing, feature refinement, class balancing, hyperparameter tuning, threshold optimization, and robustness testing under simulated label noise.

Across the evaluated models, the most practical final solution was a tuned XGBoost classifier using

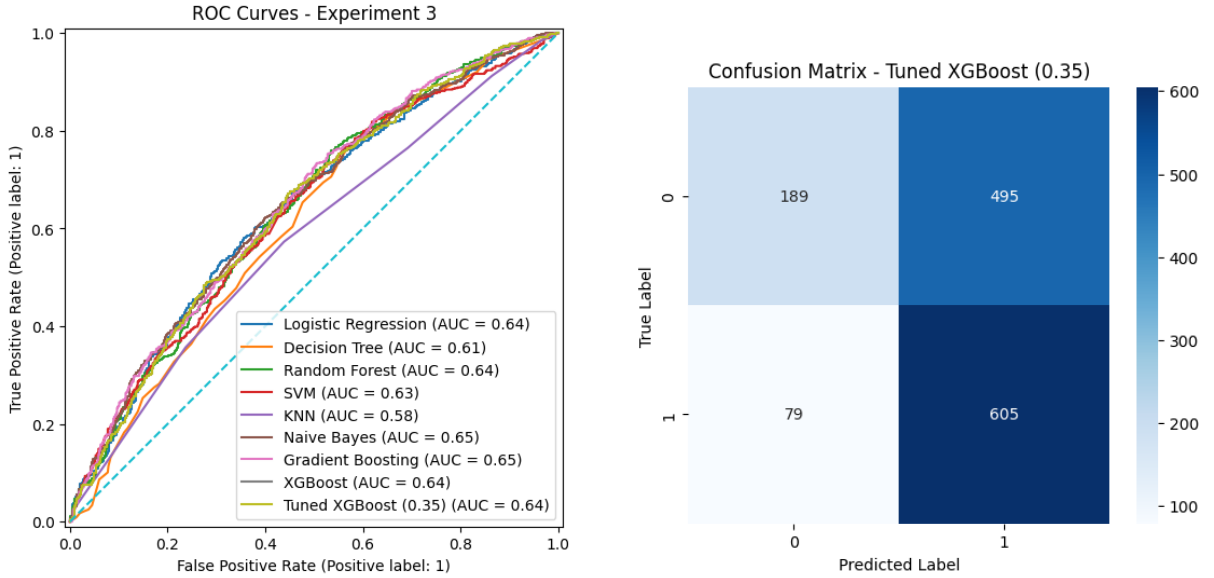

Figure 3: ROC curves for evaluated classifiers and confusion matrix of the final tuned XGBoost model at threshold = 0.35.

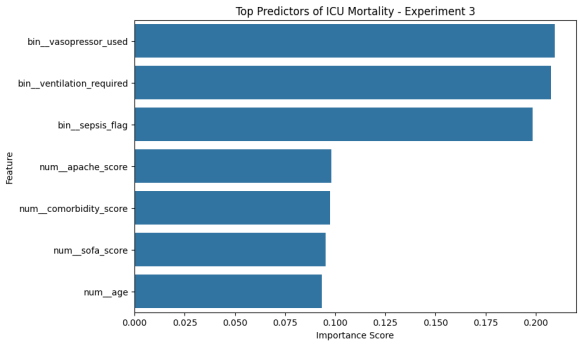

Figure 4: Relative importance of predictors in the final tuned XGBoost model.

a probability threshold of 0.35. Although some competing models achieved comparable accuracy or ROC-AUC values, the selected model provided the most suitable balance for the study objective by prioritizing mortality detection while maintaining reasonable predictive performance. Interpretability analysis further showed that vasopressor use, ventilation requirement, sepsis status, APACHE score, SOFA score, comorbidity burden, and age were the most influential predictors. These variables are clinically consistent with established indicators of critical illness severity, which increases confidence in the realism of the learned patterns.

The label noise experiment also demonstrated that data quality remains highly important in healthcare machine learning. While noisy-label training increased recall, it reduced precision and balance, highlighting the risks of unreliable documentation or imperfect labels in real clinical environments. Taken together, the findings suggest that machine learning can provide useful decision support for identifying high-risk ICU patients when model development emphasizes recall, transparency, and realistic robustness assessment. Such systems should be viewed as clinical support tools rather than replacements for professional judgment.

Future work may include external validation on multi-hospital datasets, comparison with advanced imbalance-handling methods such as SMOTE or cost-sensitive learning, probability calibration, explainable AI methods such as SHAP, and prospective evaluation within real clinical workflows.

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